Bayes' Theorem

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Bayes' Theorem (Revisted)

Introduction to Bayes' Rule

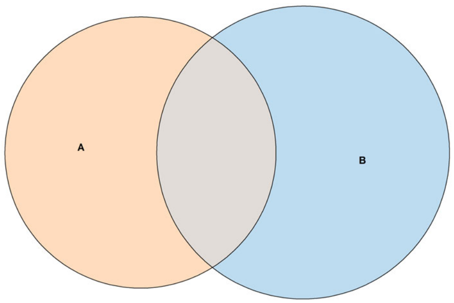
- Bayes' Rule is a foundational concept in statistics that provides a framework for updating our beliefs based on new evidence.
- It is widely used in fields such as medicine, finance, and social sciences to help make decisions under uncertainty.
- The general formula for Bayes' Rule helps to calculate posterior probabilities by combining prior knowledge and new data.

Bayes' Theorem Formula

- Bayes' Theorem allows us to "flip" conditional probabilities:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

It relates:

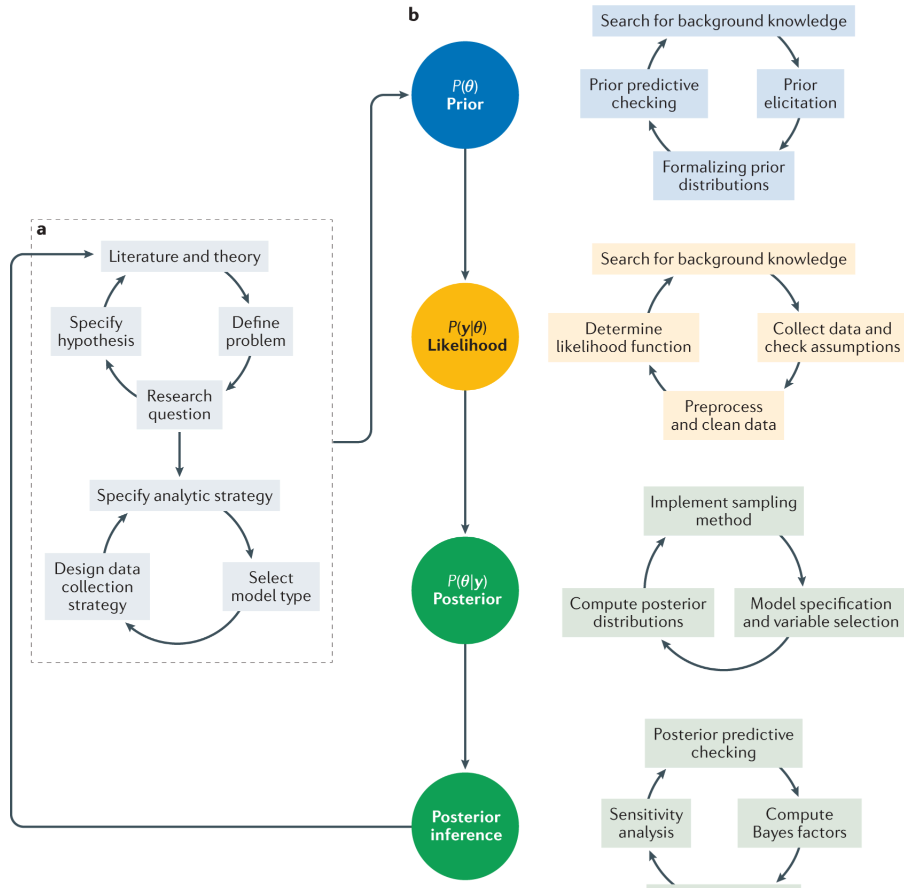


- What we want to know: P(A|B)
- What we know: P(B|A), P(A), and P(B)

Bayes' Theorem: Intuition

- Starts with our initial belief about A: P(A)
- Updates that belief based on new evidence B
- Gives us a new, updated belief: P(A|B)

This mirrors the scientific process of updating hypotheses based on evidence!



van de Schoot, R., Depaoli, S., King, R. et al. Bayesian statistics and modelling. Nat Rev Methods Primers 1, 1 (2021). `https://doi.org/10.1038/s43586-020-00001-2`

Gardner's Boy or Girl Paradox

The Paradox

Martin Gardner presented two seemingly similar problems in Scientific American (1959):

1. "Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?"
2. "Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?"

+ Gardner gave the answers as 1/2 and 1/3 respectively, leading to much debate.

Problem 1: The Older Child is a Girl

"Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?"

- Only the outcomes where the older child is a girl are considered:
  - Girl, Girl (GG)
  - Girl, Boy (GB)

P(both girls | older is girl) = 1/2

Sample space outcomes:

Option 1	**Option 2**
Girl	Girl
Girl	Boy

Option 3	**Option 4**
Boy	Girl
Boy	Boy

Problem 2: At Least One Boy

"Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?"

- Possible outcomes:
  - Boy, Boy (BB)
  - Boy, Girl (BG)
  - Girl, Boy (GB)

P(both boys | at least one boy) = 1/3

All Outcomes

Child 1	Child 2
Boy	Boy
Boy	Girl
Girl	Boy
Boy	Boy

The Paradox Explained

The key difference lies in the information given:

1. In the first problem, we know the order (older child is a girl), which eliminates two possibilities (BB and BG).
2. In the second problem, we don't know which child is a boy, so we only eliminate one possibility (GG).

This subtle difference in information leads to different sample spaces and thus different probabilities.

Bayesian Perspective

We can also approach these problems using Bayes' Theorem:

For Problem 1:

$$P(GG \mid \text{older is G}) = \frac{P(\text{older is G} \mid GG) \cdot P(GG)}{P(\text{older is G})} = \frac{1 \times \frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

For Problem 2:

$$P(BB \mid \text{at least one B}) = \frac{P(\text{at least one B} \mid BB) \cdot P(BB)}{P(\text{at least one B})} = \frac{1 \times \frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Applying Bayes' Theorem

Example: Testing for a Rare Disease

Imagine a test for a disease that affects 1% of the population:

- 99% accurate for people who have the disease (sensitivity)
- 99% accurate for people who don't have the disease (specificity)

> If someone tests positive, what's the probability they actually have the disease?

- We can use two approaches: a **tree diagram** to break down all possible outcomes and **Bayes' Theorem** mathematically.

Setting Up the Problem

Let's define our events:

- D: Having the disease (Case of the Mondays)
- T: Testing positive

We want to know: P(D|T)
We know:

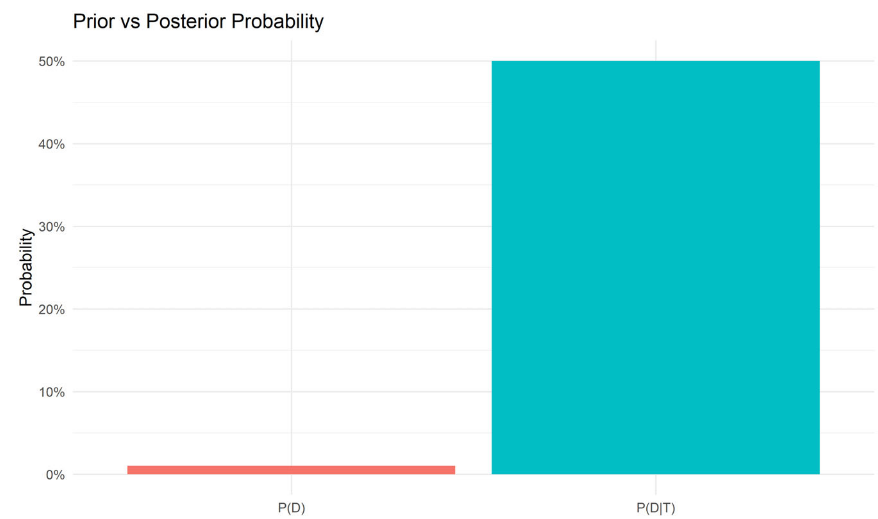
- P(D) = 0.01 (1% of population has the disease)
- P(T|D) = 0.99 (99% sensitivity)
- P(T|not D) = 0.01 (1 - 99% specificity)

Applying Bayes' Theorem mathematically

$$P(D \mid T) = \frac{P(T \mid D) \cdot P(D)}{P(T)}$$

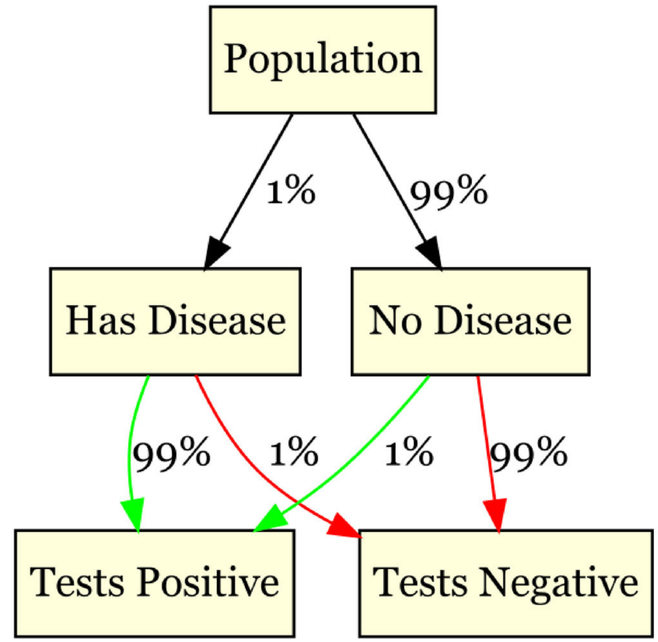
$$P(T) = P(T \mid D) \cdot P(D) + P(T \mid \neg D) \cdot P(\neg D) = 0.99 \times 0.01 + 0.01 \times 0.99 = 0.0198$$

$$P(D \mid T) = \frac{0.99 \times 0.01}{0.0198} \approx 0.50 \text{ or } 50\backslash\%$$



Applying Tree Diagrams for Rare Disease Testing

- A **tree diagram** is a graphical way to represent all possible outcomes of an event, making it easier to understand conditional probabilities.
- Let's define the population for simplicity:
  - **D**: Having the disease (1%)
  - **No D**: Not having the disease (99%)
- We will branch out from each state (having the disease or not) to show the possible test outcomes (Positive or Negative), which depend on sensitivity and specificity.



Interpretation Using the Tree Diagram

The **tree diagram** helps us visualize all possible outcomes:

- **Path 1 (Disease -> Positive Test):** Probability is **0.01 * 0.99 = 0.0099**
- **Path 2 (Disease -> Negative Test):** Probability is **0.01 * 0.01 = 0.0001**
- **Path 3 (No Disease -> Positive Test):** Probability is **0.99 * 0.01 = 0.0099**
- **Path 4 (No Disease -> Negative Test):** Probability is **0.99 * 0.99 = 0.9801**

Using Bayes' Theorem to find P(Disease | Positive Test):

$$P(D \mid T) = \frac{P(T \mid D) \cdot P(D)}{P(T)}$$

$$P(T) = P(T \mid D) \cdot P(D) + P(T \mid NotD) \cdot P(N$$

$$P(D \mid T) = \frac{0.0099}{0.0198} \approx 0.50 \text{ or } 50\backslash\%$$

Wrapping Up

- Both approaches—the tree diagram and Bayes' Theorem—help arrive at the same conclusion: there's a 50% chance that someone who tests positive actually has the disease.
- The tree diagram provides a visual breakdown, while Bayes' Theorem offers a mathematical perspective.
- This example demonstrates how rare diseases can lead to high rates of false positives, even when the test accuracy is high.

Bayesian Thinking in Psychology

Fun Example: Updating a Belief

- A pet owner initially thinks their cat is limping due to an injury (80% sure).
- After observing the cat immediately stop limping when the door opens, how should they update their belief?

$$P(\text{injury} \mid \text{cat stops limping when door opens}) =$$

$$\frac{P(\text{cat stops limping when door opens} \mid \text{injury}) \cdot P(\text{injury})}{P(\text{cat stops limping when door opens})}$$

Research Example: P-hacking

- P-hacking: Trying multiple analyses until you get a "significant" p-value
- Bayesian perspective: This is like ignoring your prior probabilities and only focusing on P(data | hypothesis)
- Better approach: Consider P(hypothesis) and calculate P(hypothesis | data)

Pros and Cons of Bayesian Approaches

Pros:

- Allows incorporation of prior knowledge
- Provides direct probabilities of hypotheses
- Handles small samples and rare events well

Cons:

- Selecting priors can be subjective
- Computationally intensive for complex problems
- Not as widely used (yet) in psychology

Wrapping Up...

Simulations

- Simulations are used to understand probabilities in complex situations.
- **Blackjack**: Understanding dealer's odds.
  - `Dealer's Odds in Blackjack`
- **Rock Paper Scissors**: Using AI Player.
  - `Rock Paper Scissors with AI Player`
- **Texas Hold'em**: Exploring probabilities of winning.
  - `Probabilities of Winning in Texas Hold'em`
- **Lottery (Adam Lamers)**: Running simulations to determine lottery odds.
  - `Lottery Simulator by Adam Lamers`